CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Tech. Examination (EC/ME/CL)

December 2011

EE 102 Fundamentals of Electrical Engineering

Date: 08.12.2011, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures where ever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Define the following: **[07]**

a) Magneto motive force
b) Magnetic Flux Density
c) Mutual Inductance
d) Leakage Coefficient
e) Kirchhoff's current law
f) Lenz's law
g) Coulomb's second law

Q - 2 (a) State Faraday's first and second law. **[02]**

(b) Derive expression of equivalent inductance when two magnetically coupled coils are connected in series in two different ways. **[07]**

(c) Two capacitors having capacitances of 20μF and 30μF are connected in series across a 600V dc supply. Calculate the potential difference across each capacitor. If a third capacitor of unknown capacitance is now connected in parallel with the 20μF capacitor such that the potential difference across 30μF capacitor is 400V. Calculate (i) the value of unknown capacitance (ii) energy stored in the third capacitor. **[05]**

OR

Q - 2 (a) State the similarities and dissimilarities between electric and magnetic circuits. **[07]**

(b) An iron ring of mean diameter 19.1 cm and having a cross sectional area of 4 cm^2 is required to produce a flux of 0.44 mWb. Find the ampere turns to be provided on the coil wound on the ring. If a saw cut of 1mm wide is made in the above ring, how many ampere turns are required? The points on B-μ_r curve are given below: **[05]**

B (T)	0.8	1.0	1.2	1.4
μ_r	2300	2000	1600	1100

(c) The resistance of a given conductor is 50Ω at 20°C. Find its temperature coefficient and resistance at 35°C (take $\alpha_0 = 0.0043°C^{-1}$). **[02]**

Q - 3 (a) A 10 μF capacitor in series with 1MΩ resistor is connected across a 100 V dc supply. Determine:- **[07]**

(i) Time constant of the circuit
(ii) The initial value of charging current

(iii) The initial rate of rise of voltage across the capacitor
(iv) Voltage across the capacitor after a time equal to time constant.
(v) The circuit current at this time
(vi) The voltage across the capacitor 3 second after closing the switch
(vii) The time taken for the capacitor voltage to reach 60 V.

(b) A network of resistance is formed as follows: **[03]**
(i) Resistance between A and B = 36Ω
(ii) Resistance between B and C = 18Ω
(iii) Resistance between C and A = 27Ω
(iv) Resistance between A and D = 12Ω
(v) Resistance between B and D = 08Ω
(vi) Resistance between C and D = 06Ω
Compute the network resistance between 1) A and B 2)B and C 3) C and A.

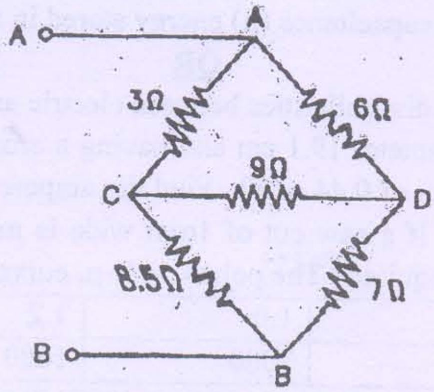
(c) A parallel plate capacitor has plates of area $2m^2$ spaced by the three slabs of different dielectrics. The relative permittivity's are 2, 3 and 6 and the thicknesses are 0.4, 0.6 and 1.2 mm respectively. Calculate the combined capacitance and the dielectric stress in each material when the applied voltage is 1000V. **[04]**

OR

Q - 3 (a) Show that if α_{t1} is the temperature coefficient of resistance of a conductor at t_1 °C, the temperature coefficient of resistance at t_2 °C is given by **[08]**

$$\alpha_{t2} = \frac{1}{\frac{1}{\alpha_{t1}} + (t_2 - t_1)}$$

(b) Using delta/star transformation, find the value of the equivalent resistance between points A and B. **[06]**



SECTION – II

Q - 4 (a) State the relations for the following: **[02]**
a) Between frequency and time period
b) Between angular velocity and frequency

(b) Derive an expression for the instantaneous value of alternating sinusoidal emf in terms of its maximum value, angular frequency and time. **[05]**

- 5 (a) For the below given cases what will be the value of power factor when power measurement is done in 3-Ø system by two wattmeter's W_1 and W_2: [05]

a) Reading of $W_1 = W_2$
b) Reading of $W_1 = - W_2$
c) Reading of $W_1 = 0W$ and $W_2 = 10W$
d) Reading of $W_1 > W_2$
e) Reading of $|W_1| > |W_2|$

(b) Draw figures & label them properly as per the requirement of the following cases: [04]

a) Draw the circuit diagram & phasor diagram for inductive load connected to a 3-Ø system whose power has to be measured by 2 wattmeter method.
b) Draw the circuit diagram & phasor diagram for capacitive load connected to a 3-Ø system whose power has to be measured by 2 wattmeter method.

(c) A 3-Ø 400V, 50 Hz induction motor has full load output of 14.9kW at which the efficiency and power factor are 0.88 and 0.8 respectively. Find the reading of two wattmeter connected to measure the power input to the motor. What is the full load line current? [03]

(d) Define the following: [02]

a) Line current
b) Phase Voltage

OR

Q - 5 (a) A coil of resistance 5Ω and inductive reactance 40Ω is connected in parallel. With a resistance of 16Ω and capacitive reactance of 20Ω. The parallel circuit is connected across 200V mains. Find the following: (a) The current in each branch. (b) Power factor of the circuit (c) Power consumed by the circuit (d) Draw the vector diagram. [05]

(b) Compare series and parallel resonance. [03]

(c) Derive expression for impedance, current and power factor for an R-C Series circuit when supplied with ac voltage. Draw also the phasor diagram. [06]

Q - 6 (a) Two Coils A & B are connected in series across a 240V, 50Hz supply. The resistance of A is 5Ω and inductance of B is 0.02H. If the input from the supply is 3kW and 2kVAR, find the inductance of A and resistance of B. Calculate voltage drop across each coil. [04]

(b) An inductive coil having a resistance of 20Ω and an inductance of 20μH is connected in parallel with a variable capacitor. This parallel combination is connected in series with a resistance of 8000Ω. A voltage of 230V at frequency of 10^6 Hz is applied across the circuit. Calculate the following: (a) Value of capacitance at resonance (b) Q factor (c) Dynamic impedance (d) Total circuit current [04]

(c) Fill in the blanks: [06]

a) Power factor of a purely capacitive circuit is ______.
b) In an inductive circuit when the frequency of applied voltage is increased, the current will _____.
c) Power drawn by three phase balanced load is given by ______.
d) Two admittances (4+j3) S and (4-j3) S are connected in parallel. The admittance of parallel combination is ______.

e) Polar form of 5+j8.66 is ______.

f) In a series RLC circuit the condition for resonance is ______.

OR

Q - 6 (a) A series RC circuit consumes a power of 7kW when connected across 200V, 50 Hz supply. The voltage across resistor is 130V. Calculate (i) Resistance, capacitance and impedance (ii) Current (iii) Equations for voltage and current **[07]**

(b) Define the following: **[04]**

a) Power factor

b) Q factor

c) Instantaneous value

d) Apparent Power

(c) Two vectors A1 = 12+j15 and A2 = 6+j8 are given. Calculate the following: **[03]**

a) Convert both the vectors in polar form.

b) Divide both the vectors

c) Multiply both the vectors

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